

Fast matrix multiplication using CP decomposition

Computational laboratory project report

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Abstract

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1 Notation and tensor preliminaries

A tensor is a multidimensional array, and in this work we deal only with 3-dimensional (or 3-way) tensors $\mathcal{X} \in \mathbb{R}^{m \times n \times p}$.

2 Low rank tensor decomposition

3 Fast matrix multiplication

Matrix multiplication is a bilinear operation. We indicate with $\langle m, n, p \rangle$ the bilinear form that represents the multiplication between a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $\mathbf{B} \in \mathbb{R}^{n \times p}$. Fixing the natural ordering on the entries of $\mathbf{A}$ and $\mathbf{B}$ and the inverse ordering on $\mathbf{C} = \mathbf{AB}$, we get a representation of $\mathcal{X} \in \mathbb{R}^{mn \times np \times mp}$ of the matrix multiplication $\langle m, n, p \rangle$ such that

$$\text{vec}(\mathbf{C}^\top) = \mathcal{X} \times_1 \text{vec}(\mathbf{A})^\top \times_2 \text{vec}(\mathbf{B})^\top. \quad (1)$$

Tensor definition

If we are given a CP decomposition of rank r of the tensor $\mathcal{X} = \llbracket U, V, W \rrbracket$, we can rewrite Equation 1 into

$$\text{vec}(\mathbf{C}^\top) = \sum_{l=1}^r (\mathbf{u}_l^\top \text{vec}(\mathbf{A})) \cdot (\mathbf{v}_l^\top \text{vec}(\mathbf{B})) \mathbf{w}_l, \quad (2)$$

where $\mathbf{u}_l, \mathbf{v}_l, \mathbf{w}_l$ denote the columns of $\mathbf{U}, \mathbf{V}$ and $\mathbf{W}$.

When $m, n, p > 2$ we can always partition them in blocks as:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix},$$

where $A_{11} \in \mathbb{R}^{\lfloor \frac{m}{2} \rfloor \times \lfloor \frac{n}{2} \rfloor}$, $B_{11} \in \mathbb{R}^{\lfloor \frac{n}{2} \rfloor \times \lfloor \frac{p}{2} \rfloor}$, $A_{22} \in \mathbb{R}^{\lceil \frac{m}{2} \rceil \times \lceil \frac{n}{2} \rceil}$, and $B_{22} \in \mathbb{R}^{\lceil \frac{n}{2} \rceil \times \lceil \frac{p}{2} \rceil}$.

In particular, we can view the product AB as the product between 2×2 block matrices

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \cdot \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

The naive algorithm proceeds by combining a set of eight matrix multiplications with four additions:

$$\begin{aligned} M_1 &= A_{11} \cdot B_{11} & M_2 &= A_{12} \cdot B_{21} & M_3 &= A_{11} \cdot B_{12} \\ M_4 &= A_{12} \cdot B_{22} & M_5 &= A_{21} \cdot B_{11} & M_6 &= A_{22} \cdot B_{21} \\ M_7 &= A_{21} \cdot B_{12} & M_8 &= A_{22} \cdot B_{22} \end{aligned}$$

$$\begin{aligned} C_{11} &= M_1 + M_2 & C_{12} &= M_3 + M_4 \\ C_{21} &= M_5 + M_6 & C_{22} &= M_7 + M_8 \end{aligned}$$

References